

MNIST_Digit_Classification_with_Logistic_Regression

Some Helper Function:

```
Dataset Preview:
   7  0  0.1  0.2  0.3  0.4  0.5  0.6  0.7  0.8  ...  0.658  0.659  0.660  \
0  2  0    0    0    0    0    0    0    0    0  ...    0    0    0
1  1  0    0    0    0    0    0    0    0    0  ...    0    0    0
2  0  0    0    0    0    0    0    0    0    0  ...    0    0    0
3  4  0    0    0    0    0    0    0    0    0  ...    0    0    0
4  1  0    0    0    0    0    0    0    0    0  ...    0    0    0

   0.661  0.662  0.663  0.664  0.665  0.666  0.667
0    0    0    0    0    0    0    0
1    0    0    0    0    0    0    0
2    0    0    0    0    0    0    0
3    0    0    0    0    0    0    0
4    0    0    0    0    0    0    0

[5 rows x 785 columns]

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9999 entries, 0 to 9998
Columns: 785 entries, 7 to 0.667
dtypes: int64(785)
memory usage: 59.9 MB
None
```

Shapes:

```
X_train: (7999, 784) y_train: (7999,)
X_test: (2000, 784) y_test: (2000,)
```

Softmax Test Case:

```
Softmax function passed the test case!
```

Test Function for Prediction Function:

```
Predicted class labels: [1 1 0]
```

Test case for Loss Function:

```
Cross-Entropy Loss (Correct Predictions): 0.4304
Cross-Entropy Loss (Incorrect Predictions): 8.9872
```

Test Case for Cost Function:

```
Cost for correct prediction: 0.0006234364133349324  
Cost for incorrect prediction: 0.29930861359446115  
Test passed!
```

Test case for compute_gradient function:

```
Gradient w.r.t. W: [[ 0.1031051  0.01805685 -0.12116196]  
 [-0.13600547  0.00679023  0.12921524]]  
Gradient w.r.t. b: [-0.03290036  0.02484708  0.00805328]  
Test passed!
```